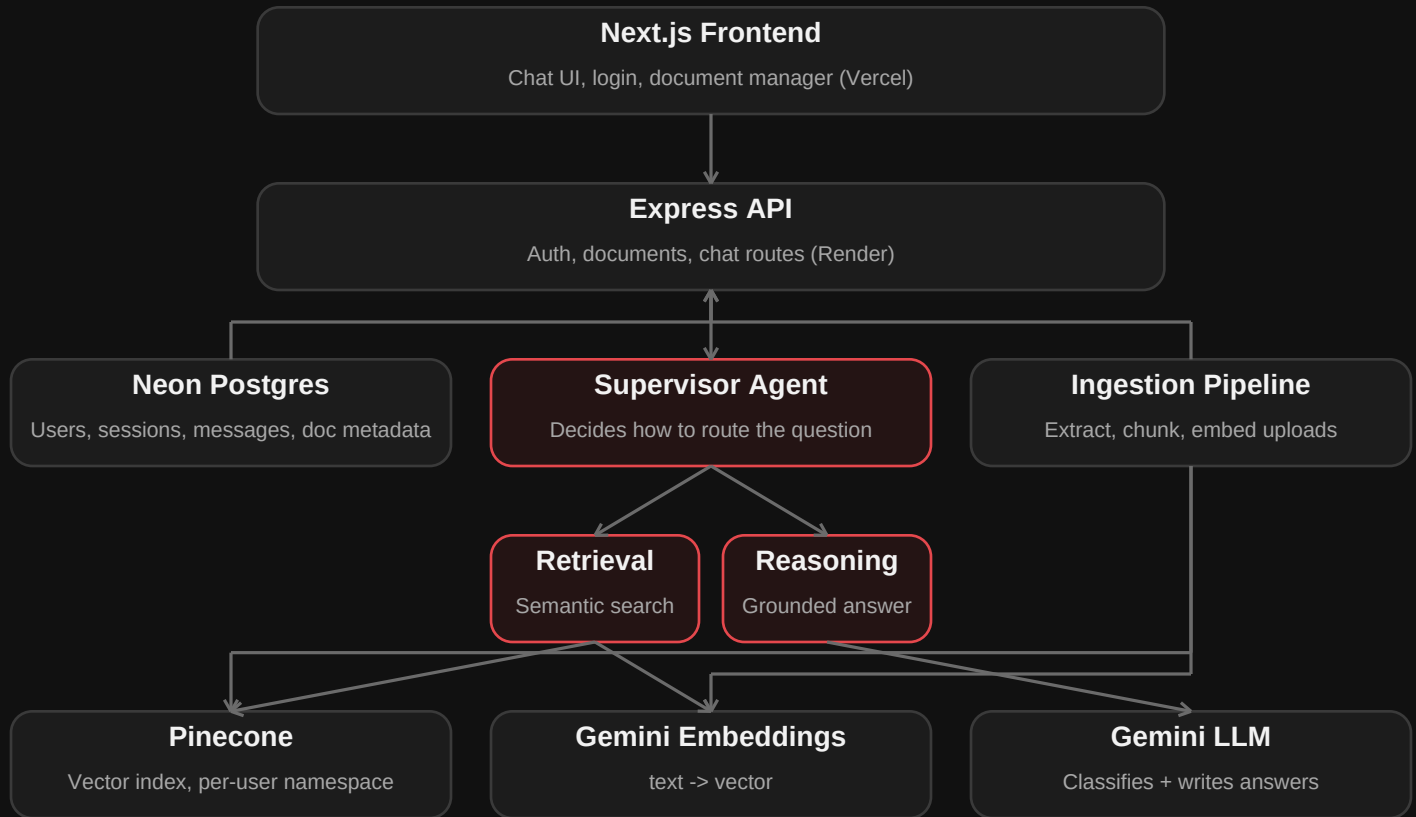


Multi-Agent RAG Chatbot

System Architecture Overview



Red-outlined boxes = the LangGraph multi-agent core (Supervisor -> Retrieval -> Reasoning)

How It Works, In Plain English

This app lets you upload your own documents (PDF, TXT, or Word files) and then ask questions about them in a chat window. Instead of one AI doing everything, the work is split between three specialist "agents" that hand off to each other, plus a small set of supporting services.

The three agents

- **Supervisor** reads your question first and decides what kind of question it is: a question about your uploaded documents, a question about the conversation itself (like "what did I just ask?"), or just general chit-chat.
- **Retrieval** only runs when the Supervisor decides your question is about your documents. It searches through everything you've uploaded and pulls out the most relevant passages.
- **Reasoning** always runs last. It takes whatever passages Retrieval found (if any), plus your conversation history, and writes the actual answer — including citations back to the source document so you can trust where the answer came from.

The supporting pieces

- **Next.js Frontend** is the web page you actually use — the chat window, login screen, and document manager. It's hosted on Vercel.
- **Express API** is the backend server that receives your messages, checks who you are, and runs the three agents. It's hosted on Render.
- **Neon Postgres** is the database that remembers your account, your chat sessions, every message you've sent, and which documents belong to you.
- **Pinecone** is a specialized database built for "meaning-based" search. Instead of matching exact keywords, it finds passages that are conceptually similar to your question.
- **Gemini** (Google's AI) plays two roles: it converts text into the numeric "fingerprints" (embeddings) that Pinecone searches with, and it's the actual language model that classifies questions and writes the final answers.

What happens when you upload a document

Your file is read and its text is pulled out, then split into small overlapping chunks (about a paragraph each). Every chunk is converted into an embedding by Gemini and stored in Pinecone, tagged with your user ID so only you can ever retrieve it. A record of the document and its chunks is also saved in Postgres so citations can show the exact source text later.

What happens when you ask a question

- 1 Your message and your recent conversation history are sent to the Supervisor.
- 2 The Supervisor picks a route: document question, history question, or general chat.
- 3 If it's a document question, Retrieval turns your question into an embedding, searches Pinecone for the closest matching chunks, and hands them to Reasoning.
- 4 Reasoning writes the final answer using only what it was actually given — if the documents don't contain the answer, it says so instead of guessing.
- 5 The question and answer are saved to Postgres, and the answer (with any citations) is sent back to the chat window.